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WORKBOOK

Memory — Workbook

A Brain Health learning product

Foundational • General public

Memory - Workbook

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Memory: An Introductory Workbook

Understanding How We Learn, Remember, Forget, and Improve Recall

Workbook Purpose

Memory is not a perfect recording of the past. It is an active, biological, and psychological process that helps us use experience to guide present and future action.

This workbook introduces the basics of memory for a general audience. You will learn:

- What memory is and why it matters
 - How memories are formed, stored, and retrieved
 - Why forgetting happens
 - Why memory can be inaccurate
 - How attention, sleep, emotion, and practice affect memory
 - Practical strategies for remembering more effectively
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How to Use This Workbook

You can read this workbook straight through or complete it section by section. Throughout, you will find:

- Key ideas
- Everyday analogies
- Reflection prompts
- Practice activities
- Quick checks
- Memorable takeaways

You may want a notebook or digital document for your answers.

1. What Is Memory?

Definition

Memory is the set of mental and brain processes that allow us to encode, store, and retrieve information.

In everyday terms, memory helps us:

- Recognize familiar people and places
- Use language
- Learn skills
- Make decisions
- Maintain a sense of personal identity
- Plan for the future
- Avoid repeated mistakes

Memory is not one single ability. It includes many systems working together.

The Three Basic Stages of Memory

Most memory processes involve three broad stages:

Stage	What It Means	Example
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Encoding	Taking in information and converting it into a form the brain can use	Hearing someone's name and paying attention to it
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Storage	Maintaining information over time	Keeping that name available in memory
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Retrieval	Accessing stored information when needed	Remembering the name later
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Everyday Analogy: Memory as a Workshop

Think of memory like a busy workshop.

- Encoding is bringing materials into the workshop.
- Storage is organizing those materials on shelves.
- Retrieval is finding the right tool or part when you need it.

If materials come in too quickly, are poorly labeled, or are stored randomly, they may be hard to find later.

Reflection Prompt

Think of a time when you forgot something important, such as a name, appointment, password, or task.

1. What did you forget?
 2. Do you think the problem was with encoding, storage, or retrieval?
 3. What could have helped?
-

Memorable Takeaway

Memory is not simply "having information." It is the process of getting information in, keeping it available, and finding it again when needed.

2. The Main Types of Memory

Memory is often divided into several types based on how long information lasts and what kind of information is involved.

Sensory Memory

Sensory memory briefly holds information from the senses.

Examples:

- A visual image lingering for a fraction of a second
- The echo of a sound lasting briefly after it ends

Sensory memory is very short-lived, but it helps create a continuous experience of the world.

Working Memory

Working memory is the mental workspace where we hold and use information temporarily.

Examples:

- Remembering a phone number long enough to dial it

- Doing mental arithmetic
- Following directions
- Keeping track of a conversation

Working memory has limited capacity. It is easily disrupted by distraction.

Long-Term Memory

Long-term memory stores information over longer periods, from minutes to a lifetime.

Examples:

- Your childhood memories
- Facts you know
- Skills like riding a bicycle
- Vocabulary
- Personal experiences

Long-term memory is large, but not always perfectly accurate or easy to retrieve.

Quick Comparison

Type of Memory	Duration	Capacity	Example
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Sensory memory	Very brief	Large but fleeting	Seeing a sparkler trail
Working memory	Seconds to minutes	Limited	Holding a number in mind
Long-term memory	Minutes to lifetime	Very large	Knowing your address

Practice Activity: Working Memory Test

Read the following numbers once, then look away and try to write them down in order:

7 - 2 - 9 - 4 - 1 - 8 - 6

Now try this one:

4 - 9 - 2 - 7 - 5 - 1 - 8 - 3 - 6 - 0

Reflection

1. Which sequence was easier?
 2. Did you use any strategy?
 3. What happened when the list got longer?
-

Memorable Takeaway

Working memory is like a small mental desktop: useful, but easily cluttered.

3. Long-Term Memory: What We Remember

Long-term memory can be divided into different categories.

Explicit Memory

Explicit memory involves information we can consciously recall and describe.

There are two major forms:

Episodic Memory

Memory for personal experiences.

Examples:

- Your last birthday
- A vacation
- A conversation you had yesterday

Semantic Memory

Memory for facts, concepts, and general knowledge.

Examples:

- Paris is the capital of France
 - A triangle has three sides
 - Dogs are mammals
-

Implicit Memory

Implicit memory influences behavior without requiring conscious recall.

Examples:

- Riding a bike
- Typing on a keyboard
- Recognizing familiar patterns
- Feeling uneasy in a place because of a past experience you may not clearly remember

One important type of implicit memory is procedural memory, which supports learned skills and habits.

Memory Types at a Glance

Memory Type	Description	Example
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Episodic	Personal experiences	Remembering your first day at school
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Semantic	Facts and concepts	Knowing the meaning of a word
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Procedural	Skills and habits	Tying your shoes
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Emotional associations	Learned emotional responses	Feeling nervous near a place linked to stress
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Everyday Analogy: Different Files, Different Uses

Long-term memory is not one filing cabinet. It is more like a library with different sections:

- Autobiographies for life events
- Encyclopedias for facts
- Instruction manuals for skills
- Warning labels for emotional learning

Each section stores information differently and may be accessed in different ways.

Reflection Prompt

List one example from your own life for each memory type:

Memory Type	Your Example
-------------	--------------

|---|---|

| Episodic memory | |

| Semantic memory | |

| Procedural memory | |

| Emotional association | |

Memorable Takeaway

Memory includes more than remembered events. It also includes facts, skills, habits, and emotional learning.

4. How Memories Are Formed

Memory formation depends on several processes, especially attention, meaning, repetition, and consolidation.

Attention: The Gateway to Memory

We are exposed to far more information than we can remember. Attention helps select what gets encoded.

If you do not pay attention to something, it may never become a strong memory.

Example:

You may have seen hundreds of coins, but you might still struggle to draw one accurately from memory because you rarely attend to the details.

Meaning Matters

Information is easier to remember when it connects to what you already know.

For example, the letters:

C A T

are easier to remember than:

T C A

because "cat" has meaning.

Repetition Helps, But Not All Repetition Is Equal

Simply repeating information can help, but deeper practice is usually better.

| Weak Strategy | Stronger Strategy |

|---|---|

| Rereading the same page many times | Testing yourself on the material |

| Repeating a word without meaning | Using the word in a sentence |

| Cramming the night before | Reviewing over several days |

| Highlighting everything | Identifying key ideas and explaining them |

Consolidation

Consolidation is the process by which memories become more stable over time.

Sleep plays an important role in consolidation. During sleep, the brain helps strengthen and organize memories.

The Brain and Memory: A Simple Introduction

Memory involves many brain regions. Two especially important areas are:

Hippocampus

The hippocampus helps form new explicit memories, especially memories for events and facts.

Amygdala

The amygdala is involved in emotional learning and helps emotional events become memorable.

Memory is not stored in one single "memory center." It depends on networks across the brain.

Practice Activity: Make It Meaningful

Try to remember this list:

- Apple
- Train
- Candle
- River
- Jacket
- Window

Now create a short story using all six words.

Example:

"An apple rolled onto a train seat beside a candle, while outside the window a river flowed past a person wearing a red jacket."

After a few minutes, cover the list and write as many words as you can remember.

Reflection

1. Did the story help?
2. Why might meaning improve memory?
3. How could you use this method in daily life?

Memorable Takeaway

We remember best when we pay attention, connect new information to meaning, and revisit it over time.

5. Retrieval: Finding What You Know

Sometimes information is stored, but we cannot access it when we want to. This is a retrieval problem.

Retrieval Cues

A retrieval cue is a clue that helps bring a memory back.

Examples:

- A smell that reminds you of a place

- A song that brings back a time in your life
 - Seeing a person's face and remembering their name
 - Returning to a room and remembering why you went there
-

Context Matters

Memory is often easier when the retrieval context matches the learning context.

Examples:

- You remember a conversation better when you are back in the same location.
 - A student may recall material better when tested in a similar setting to where they studied.
 - A familiar smell or sound may trigger a memory.
-

The Tip-of-the-Tongue Experience

The tip-of-the-tongue state occurs when you feel that you know something but cannot retrieve it immediately.

Common signs:

- You may remember the first letter.
- You may remember related words.
- The answer may come later when you stop trying so hard.

This shows that memory retrieval can be partial.

Practice Activity: Retrieval Cues

Think of a song from your teenage years or an earlier period of life.

1. What memory does it bring back?
 2. Is the memory visual, emotional, social, or location-based?
 3. Why do you think music can be such a strong cue?
-

Memorable Takeaway

Forgetting is not always losing a memory. Sometimes it is losing the path to it.

6. Why We Forget

Forgetting is normal. It does not always mean something is wrong.

Common Reasons for Forgetting

1. Encoding Failure

The information was never strongly encoded.

Example:

You forget where you put your keys because you were distracted when you set them down.

2. Decay

Some memories weaken over time, especially if they are not used.

Example:

You forget details from a class you took years ago.

3. Interference

Other information competes with what you are trying to remember.

There are two common forms:

Type	Description	Example
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Proactive interference	Old information interferes with new information	Your old password makes it harder to remember your new one
Retroactive interference	New information interferes with old information	Your new phone number makes it harder to recall your old one

4. Retrieval Failure

The memory may exist, but you cannot access it at the moment.

Example:

You recognize someone but cannot remember their name.

5. Motivated Forgetting

People may avoid thinking about painful, embarrassing, or threatening experiences. This area is complex, and memory for trauma varies widely. Some people remember traumatic events vividly; others have fragmented or incomplete recall.

Everyday Analogy: A Path Through Grass

Imagine walking the same path through grass every day. The path becomes clear and easy to follow.

If you stop using it, grass grows back and the path becomes harder to see.

Memory works somewhat similarly: use strengthens access; disuse can weaken it.

Reflection Prompt

Choose one thing you often forget.

Examples:

- Names
- Passwords
- Appointments
- Where you put objects
- Items at the store

Which reason for forgetting is most likely involved?

| What I Forget | Likely Cause | Possible Solution |

|---|---|---|

||||

Memorable Takeaway

Forgetting is not a single problem. It can happen because information was never encoded, has weakened, is blocked by other information, or is hard to retrieve.

7. Memory Is Reconstructive

One of the most important facts about memory is that it is reconstructive.

This means we do not replay the past exactly like a video recording. Instead, we rebuild memories using stored details, expectations, emotions, and current knowledge.

Why Memory Can Change

Memories can be influenced by:

- Later conversations
 - Photographs
 - Suggestions from others
 - Current beliefs
 - Emotional state
 - Repeated retelling
 - Similar events blending together
-

Example: Remembering a Conversation

You may remember the main point of a conversation but not the exact words. Later, your brain may fill in likely wording based on what you believe happened.

This is usually useful. Memory does not need perfect detail for everyday life. But it can create errors.

False Memories

A false memory is a memory of something that did not happen or happened differently from how it is remembered.

False memories do not always mean someone is lying. A person can sincerely remember

something inaccurately.

Eyewitness Memory

Eyewitness memory can be affected by:

- Stress
- Weapon focus
- Poor lighting
- Cross-racial identification difficulties
- Leading questions
- Time delay
- Exposure to misleading information
- Confidence inflation after feedback

Because of this, legal systems increasingly recognize that eyewitness memory must be handled carefully.

Practice Activity: Memory and Wording

Read these two questions:

1. "How fast were the cars going when they hit each other?"
2. "How fast were the cars going when they smashed into each other?"

Reflection

1. Which wording suggests a more severe crash?
 2. How might that wording affect memory?
 3. Why does careful questioning matter?
-

Memorable Takeaway

Memory is not a video recording. It is a reconstruction, and reconstructions can be useful, incomplete, or mistaken.

8. Emotion and Memory

Emotion affects what we notice, how strongly memories are encoded, and how easily they return.

Emotional Events Can Be Memorable

Highly emotional events often feel vivid. This may be because emotion increases attention and activates stress-related systems that influence memory formation.

Examples:

- A joyful celebration
 - A frightening accident
 - A major public event
 - A meaningful personal achievement
-

Vivid Does Not Always Mean Accurate

Emotional memories can feel very clear, but confidence and accuracy are not the same.

People may remember the central emotional details well while misremembering peripheral details.

Example:

A person may remember being frightened during an emergency but misremember exactly who said what.

Stress and Memory

Stress has mixed effects:

- Moderate stress can sometimes sharpen attention to important information.
 - Extreme stress can impair memory, especially for details.
 - Chronic stress can negatively affect learning and recall.
-

Reflection Prompt

Think of an emotional memory.

1. What parts feel especially clear?
 2. What parts are uncertain?
 3. Has your understanding of the event changed over time?
-

Memorable Takeaway

Emotion can make memories powerful, but powerful does not always mean perfectly accurate.

9. Sleep, Health, and Memory

Memory depends on the health of the brain and body.

Sleep

Sleep supports memory in several ways:

- Helps consolidate learning
- Supports attention the next day
- Reduces mental fatigue
- Helps regulate emotion
- Supports problem-solving and integration of knowledge

Lack of sleep can impair encoding, concentration, and recall.

Exercise

Regular physical activity is associated with better brain health. Exercise can support memory indirectly by improving:

- Blood flow
- Mood
- Sleep quality

- Stress regulation
 - Overall health
-

Nutrition and Hydration

A balanced diet and adequate hydration support general brain function. There is no single "magic memory food," but overall health matters.

Stress Management

Chronic stress can interfere with attention, sleep, and learning. Helpful approaches may include:

- Regular exercise
 - Relaxation breathing
 - Social support
 - Time outdoors
 - Therapy or counseling when needed
 - Reducing avoidable overload
-

Aging and Memory

Some memory changes are common with aging, such as:

- Slower recall
- More tip-of-the-tongue experiences
- Greater difficulty multitasking
- Reduced working memory capacity

However, many abilities remain strong, including:

- Vocabulary
- General knowledge
- Emotional wisdom
- Skill-based learning
- Meaning-based understanding

Significant memory problems that interfere with daily life should be discussed with a healthcare professional.

Reflection Prompt

Rate the following from 1 to 5:

| Area | Rating: 1 = Needs Attention, 5 = Strong |

|---|---:|

| Sleep quality | |

| Stress management | |

| Physical activity | |

| Focus during learning | |

| Use of memory strategies | |

Choose one area to improve this week.

Memorable Takeaway

Memory is not only a mental skill. It is supported by sleep, health, attention, and emotional balance.

10. Practical Strategies to Improve Memory

Memory improvement is usually less about "trying harder" and more about using better methods.

Strategy 1: Pay Attention on Purpose

Before trying to remember something, pause and focus.

Example:

When putting down your keys, say to yourself:

"I am placing my keys on the kitchen counter."

This improves encoding.

Strategy 2: Use Retrieval Practice

Testing yourself strengthens memory more than simply rereading.

Examples:

- Close the book and summarize what you learned.
 - Use flashcards.
 - Practice answering questions.
 - Teach the idea to someone else.
-

Strategy 3: Space Out Learning

Spaced practice means reviewing information over time rather than cramming.

Example schedule:

- Learn today
- Review tomorrow
- Review three days later
- Review one week later
- Review two weeks later

Spacing helps strengthen long-term retention.

Strategy 4: Chunk Information

Chunking means grouping information into meaningful units.

Example:

Instead of remembering:

1 9 4 5 2 0 2 4

You might remember:

1945 and 2024

Chunking reduces the burden on working memory.

Strategy 5: Make Associations

Connect new information to something familiar.

Example:

To remember someone named Rose, imagine them holding a rose.

Strategy 6: Use Visual Imagery

Images are often easier to remember than abstract words.

Example:

To remember "milk, bread, apples," imagine a giant apple sitting on a loaf of bread floating in a pool of milk.

The stranger the image, the more memorable it may be.

Strategy 7: Create External Supports

Memory is not cheating. Humans have always used tools.

Examples:

- Calendars
- Notes
- Alarms
- Labels
- Checklists
- Pill organizers
- Habit trackers

External memory aids reduce mental load.

Strategy 8: Explain, Don't Just Repeat

If you want to remember an idea, explain it in your own words.

Ask:

- What does this mean?
- Why is it true?

- How does it connect to what I already know?
 - Can I give an example?
-

Practice Activity: Choose a Memory Strategy

Pick something you want to remember this week.

Examples:

- A person's name
- A shopping list
- A work task
- A concept from this workbook
- A new password hint
- A route or set of directions

Complete the plan:

| What I Want to Remember | Strategy I Will Use | When I Will Review It |

|---|---|---|

|||

Memorable Takeaway

Better memory comes from better encoding, better cues, and better practice-not from willpower alone.

11. Common Memory Myths

Myth 1: Memory Works Like a Camera

Reality: Memory is reconstructive. We store fragments, meanings, and associations, not perfect recordings.

Myth 2: Confidence Always Means Accuracy

Reality: People can be confident and wrong. Confidence can be influenced by feedback, repetition, and social pressure.

Myth 3: Forgetting Always Means Failure

Reality: Forgetting is normal and sometimes useful. It helps reduce clutter and prioritize relevant information.

Myth 4: Older Adults Cannot Learn New Things

Reality: Older adults can learn throughout life, though some forms of recall may become slower.

Myth 5: Rereading Is the Best Way to Study

Reality: Rereading can feel familiar, but retrieval practice and spaced repetition usually produce stronger learning.

Myth 6: There Is One Simple Trick to Perfect Memory

Reality: Memory depends on attention, meaning, practice, sleep, emotion, health, and context. No single trick solves everything.

Quick Check

Mark each statement as True or False.

1. Memory is an exact recording of events.
 2. Sleep can support memory consolidation.
 3. Working memory has limited capacity.
 4. Emotional memories are always accurate.
 5. Testing yourself can improve memory.
 6. Forgetting can happen because of retrieval failure.
-

12. Daily Memory Skills Practice

Use this section to build practical habits.

Exercise A: The Name Practice

When meeting someone new:

1. Pay attention when they say their name.
2. Repeat it naturally: "Nice to meet you, Maya."
3. Connect it to something memorable.
4. Use it once more before leaving.
5. Write it down later if appropriate.

Reflection

- Did repeating the name help?
 - Did the association feel natural?
 - Could you remember the name the next day?
-

Exercise B: The Object Placement Habit

When placing an important object somewhere, pause and say:

"I am putting my in/on ."

Example:

"I am putting my wallet in the top drawer."

This improves attention and encoding.

Exercise C: The One-Minute Review

At the end of the day, write down:

1. One thing I learned today:
2. One person I interacted with:
3. One task I completed:

4. One thing I want to remember tomorrow:

This supports reflection and memory consolidation.

Exercise D: Teach It Back

After reading a section of this workbook, close it and explain the main idea out loud or in writing.

Use this structure:

- The main idea is:
 - An example is:
 - This matters because:
-

13. Personal Memory Profile

Complete the following self-assessment.

Rate each statement from 1 to 5:

1 = Rarely true

5 = Often true

Statement	Rating
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I remember information better when I understand its meaning.	
--	--

I often forget things when I am distracted.	
---	--

I use calendars, notes, or reminders effectively.	
---	--

I remember emotional events vividly.	
--------------------------------------	--

I struggle to remember names.	
-------------------------------	--

I learn better when I test myself.	
------------------------------------	--

I sleep well enough to feel mentally clear.	
---	--

I can remember skills better than facts.	
--	--

I sometimes feel sure about a memory but later discover details were wrong.	
---	--

I benefit from reviewing information over several days.	
---	--

Reflection

Based on your ratings:

1. What is one memory strength you have?
 2. What is one memory challenge you notice?
 3. Which strategy from this workbook would help most?
 4. What is one memory habit you will practice for the next seven days?
-

14. Review: Key Concepts

Memory Is a Process

Memory involves:

1. Encoding
 2. Storage
 3. Retrieval
-

Memory Has Multiple Systems

Important types include:

- Sensory memory
 - Working memory
 - Long-term memory
 - Episodic memory
 - Semantic memory
 - Procedural memory
 - Emotional memory
-

Forgetting Has Many Causes

Common causes include:

- Encoding failure
- Decay
- Interference
- Retrieval failure

- Stress or distraction
-

Memory Is Reconstructive

Memories are rebuilt, not replayed. They can change over time.

Memory Can Be Improved

Helpful strategies include:

- Attention
 - Retrieval practice
 - Spaced review
 - Chunking
 - Association
 - Visualization
 - External reminders
 - Sleep and health support
-

15. Final Knowledge Check

Answer the following questions.

Multiple Choice

1. Which stage of memory involves getting information into the brain?

- A. Retrieval
 - B. Encoding
 - C. Forgetting
 - D. Interference
-

2. Working memory is best described as:

- A. A permanent store of all life events
-

- B. A brief mental workspace for holding and using information
 - C. A type of memory only used during sleep
 - D. An exact recording of sensory experience
-

3. Which is an example of procedural memory?

- A. Knowing the capital of Italy
 - B. Remembering your last birthday
 - C. Riding a bicycle
 - D. Recalling a phone number for ten seconds
-

4. Spaced practice means:

- A. Studying only once
 - B. Reviewing information over increasing intervals
 - C. Studying everything at the last minute
 - D. Avoiding retrieval practice
-

5. A retrieval cue is:

- A. A clue that helps access a memory
 - B. A type of brain scan
 - C. A memory that is always false
 - D. A method for erasing memories
-

Short Answer

6. Explain why memory is not like a video recording.

7. Give one example of interference.

8. Name two strategies that can improve memory.

9. Why does attention matter for memory?

10. How can sleep support memory?

16. Answer Key

Quick Check Answers

1. False
 2. True
 3. True
 4. False
 5. True
 6. True
-

Final Knowledge Check Answers

1. B - Encoding

Encoding is the process of taking in information and converting it into a usable memory form.

2. B - A brief mental workspace for holding and using information

Working memory allows us to temporarily hold and manipulate information.

3. C - Riding a bicycle

Procedural memory supports skills and habits.

4. B - Reviewing information over increasing intervals

Spaced practice strengthens long-term retention better than cramming.

5. A - A clue that helps access a memory

Retrieval cues help bring stored information back into awareness.

6. Memory is not like a video recording because it is reconstructive. We rebuild memories using stored details, meaning, emotion, expectations, and later information. This means memories can be useful but also incomplete or inaccurate.

7. Example answer:

An old password makes it harder to remember a new password. This is proactive interference because old information interferes with new learning.

8. Example answers:

- Retrieval practice
 - Spaced repetition
 - Chunking
 - Visualization
 - Paying attention
 - Using external reminders
 - Making meaningful associations
-

9. Attention matters because information that is not attended to may never be strongly encoded. Without attention, the brain may not create a durable memory.

10. Sleep supports memory by helping consolidate learning, improving attention, reducing fatigue, and supporting emotional regulation.

17. Seven-Day Memory Practice Plan

Use this plan to apply what you learned.

| Day | Practice Focus | Action |

|---|---|---|

| Day 1 | Attention | Pause when placing important objects and name the location aloud |

| Day 2 | Names | Use repetition and association when meeting or recalling people |

| Day 3 | Retrieval practice | Close your notes and recall key ideas from memory |

| Day 4 | Spaced review | Review something learned earlier in the week |

| Day 5 | Chunking | Group numbers, tasks, or ideas into meaningful units |

| Day 6 | External support | Create or improve a checklist, calendar, or reminder system |

18. Personal Action Plan

Complete the following:

My Main Memory Goal

I want to improve my ability to remember:

My Biggest Memory Challenge

The main thing that gets in the way is:

Strategies I Will Use

I will use these three strategies:

- 1.
 - 2.
 - 3.
-

My Reminder System

I will support my memory with:

- Calendar reminders
 - Written notes
 - Phone alarms
 - Checklists
 - Visual cues
 - Other:
-

My Review Plan

I will review important information:

- After one day
 - After three days
 - After one week
 - After two weeks
 - Other:
-

Final Memorable Takeaway

Memory is not a perfect record of the past. It is a living system for learning, meaning, identity, and action. To remember better, pay attention, make meaning, practice retrieval, space your learning, care for your health, and use helpful cues.

Continue your understanding at QRU:

